Managing Projects to Achieve Results

Lesson 3

WESTON COLLEGE | Mindful Education

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Study Information and Tips

This Study Text replicates the video content for the lesson

Supplementary knowledge can also be found in the information sheets and text boxes that accompany the videos, as well as in the resources in the Insights section

Activities to help you consolidate your knowledge are deliberately delivered through practice questions and exercises found in the Test and Apply sections on the VLE. These help you to further build on your knowledge and apply the learning in a way that supports your assessment

You can use this Study Text to test your understanding of the learning content at the end of each lesson. It may also be helpful to refer to it when preparing for your assessments

We recommend that you also produce your own notes in addition to the Study Text. Research has shown that handwritten notes can be very effective for learning retention

If you decide to print this Study Text, set the layout to print 4 or 6 slides per page to save on paper and ink

Project Management Methodologies and their Applications

The role of Project Management Methodologies

Structured Project Management

Agile Project Management

The PRINCE2 Framework

What is a Project Management Methodology?

Before we begin our project we'll need to choose a management methodology to guide it. The Project Management Institute defines a methodology as:

> "a system of practices, techniques, procedures and rules used by those who work in a discipline".

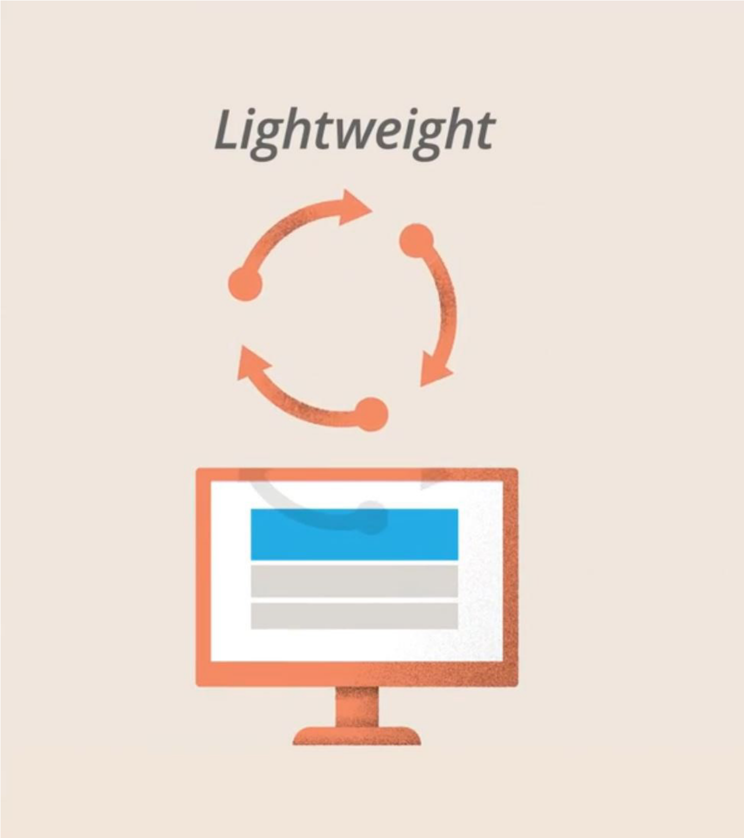
In other words, a methodology takes in a broad range of activities, but sets out quite specific ways of working.

Methodologies fall across a spectrum of approaches, from very structured to flexible, or agile.

The Range of Project Management Methodologies

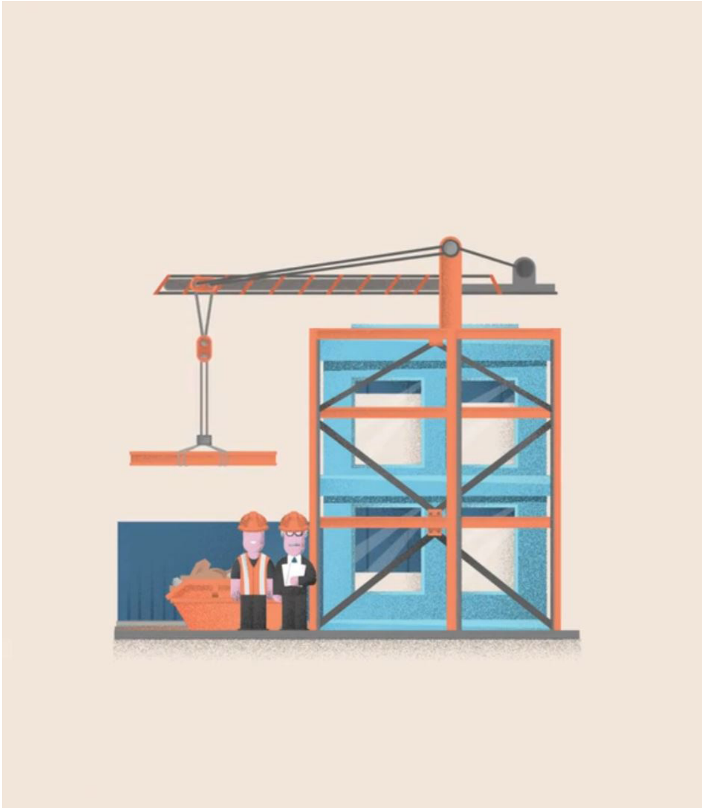
Each project methodology will be explained in detail, beginning with the structured approaches. For now, it's important to remember that one size does not fit all. Just because a method was successfully used before does not automatically mean it will work next time.

For example, if you are planning a facelift for the company website, a lightweight approach that allows you to iterate quickly is preferable. Exhaustive planning, documentation and contract stages will only increase costs, sap energy and irritate your team.



The Range of Project Management Methodologies (cont.)

If, on the other hand, you are planning to tender for a public sector construction project, you'll want a belt-and-braces methodology that covers all aspects of the process, has a rigorous planning stage, and provides clearly defined points for customer feedback. Your decision should depend primarily on the team, the customer, and, most importantly, the type of project.



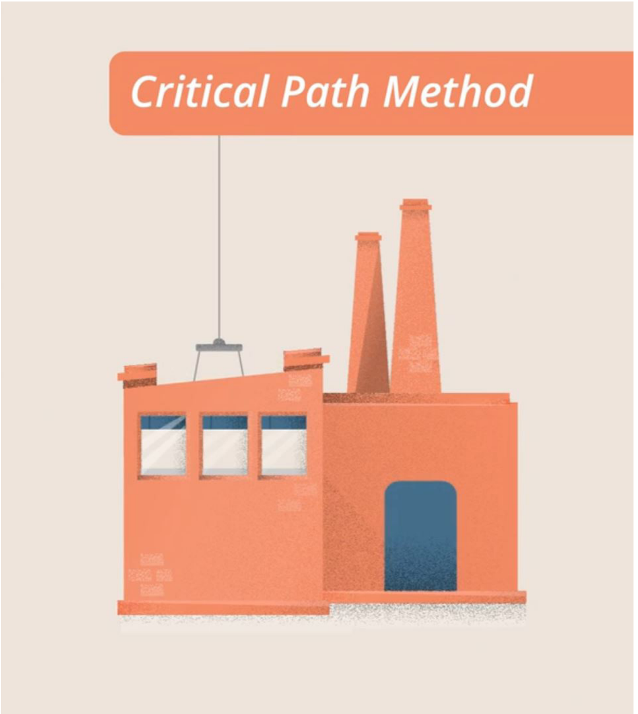
Structured and Linear Methodologies

First we will look in detail at some of the more structured and linear methodologies. These include:

- The Critical Path Method
- The Waterfall Method
- The Structures Systems Analysis and Design Method (SSADM)
- The Critical Chain Method

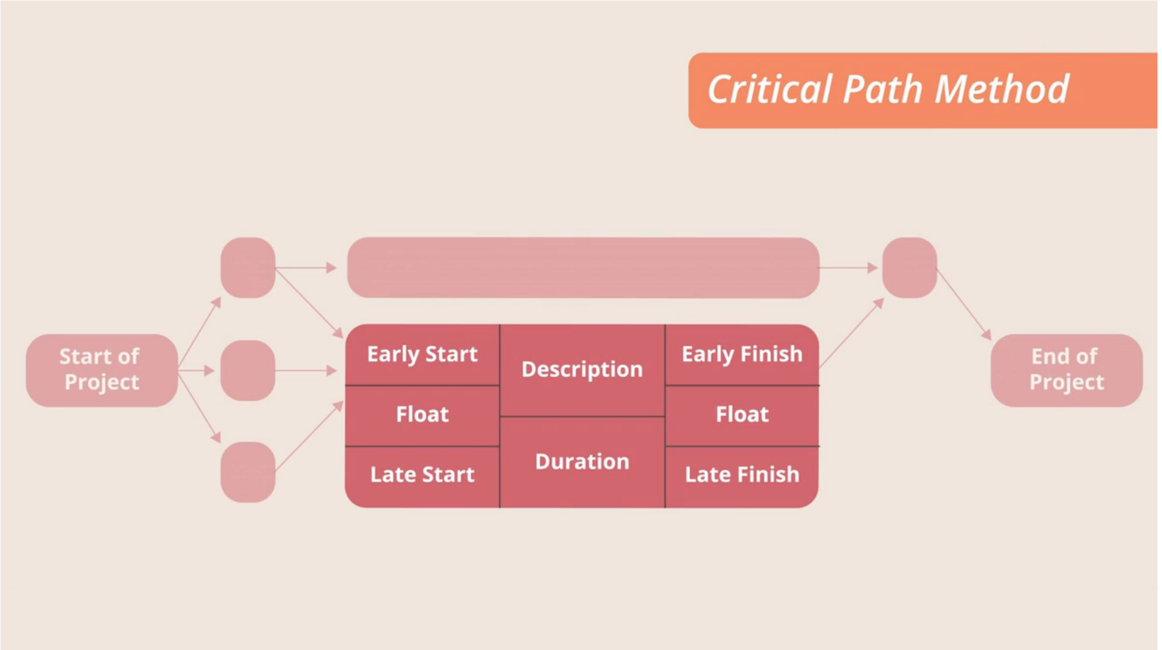
Structured and Linear Methodologies: The Critical Path Method

The Critical Path Method, or CPM was designed by engineer Morgan Walker and mathematician Jim Kelley in the 1950s, to solve the problems posed by complex engineering projects. For example, if you're building a nuclear plant, should you lower a reactor into a roofless building, or leave a hole in the wall instead? Not surprisingly, these are questions with big cost implications.



The Critical Path Method

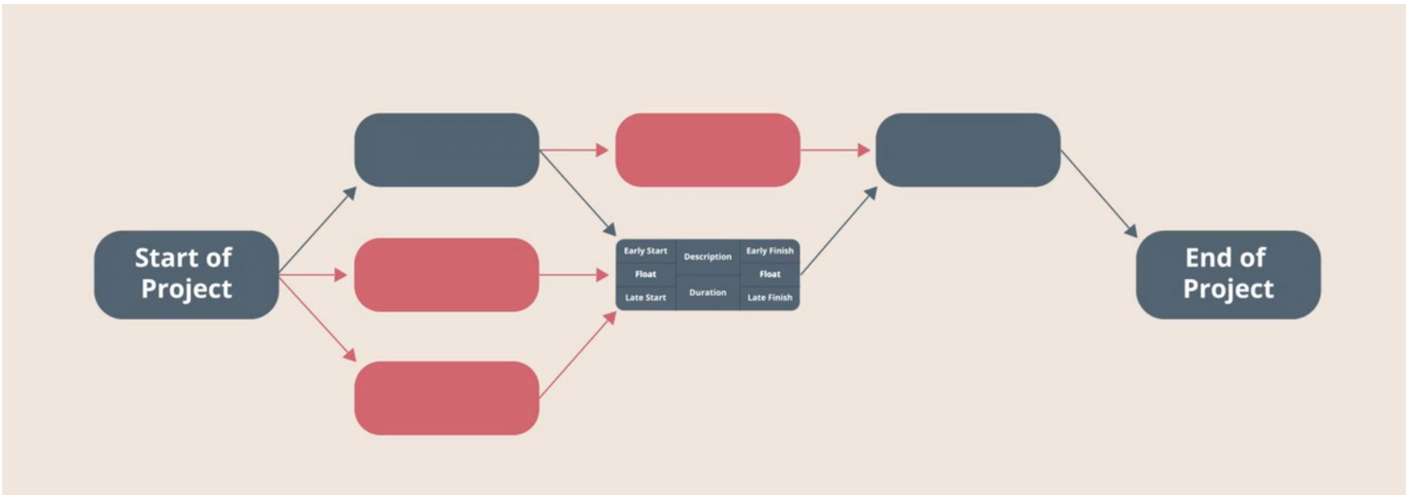
Walker and Kelley's answer was to model a project as a network of nodes and connections. Each task in the project is represented as a node, which has a description of the activity and the earliest and latest times it can start and finish. Also included is 'float time', which is the number of days the task can be delayed without delaying the overall project.



The Critical Path Method (cont.)

The critical path is the core sequence of activities that have to be completed on time, one after another, for the project to hit its deadlines.

CPM also represents tasks that can happen in parallel to the core tasks, and shows where they can rejoin the main critical path.



The Critical Path Method (cont.)

Note that CPM is sometimes used interchangeably with the PERT method, or the Program Evaluation and Review Technique. The latter was also developed in 1950s America, initially to support the Polaris nuclear missile programme. The difference is that CPM builds in cost estimates, whereas PERT has a greater emphasis on time and can factor in time estimates for various scenarios.

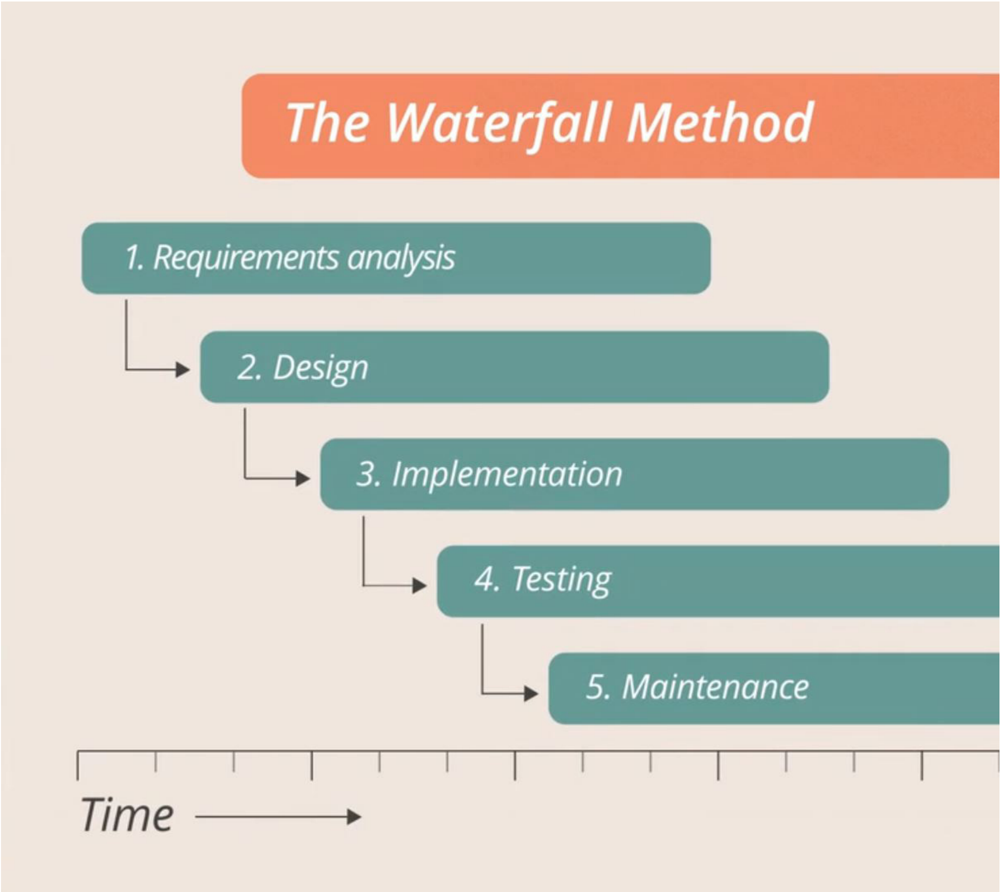
The Waterfall Method

The Waterfall method is a sequential process used across many industries. It usually contains five phases executed in a specific order. These are:

- requirements analysis
- design
- implementation
- testing
- maintenance

The Waterfall Method (cont.)

The vertical layout of these phases when plotted against time gives rise to the name Waterfall. Waterfall diagrams are familiar to many of us and extremely intuitive. They are especially useful for large-scale projects where sign-off processes require clearly identifiable phases. But the linearity of the method has invited criticism.



The Waterfall Method (cont.)

In 1970, computer scientist Dr Winston Royce described it as ‘risky and inviting failure’. Royce argued that delaying the ‘testing’ phase until the end inevitably leads to expensive, full-scale redesigns. There are various modern offshoots of the method that attempt to reduce some of the linearity, such as SSADM.

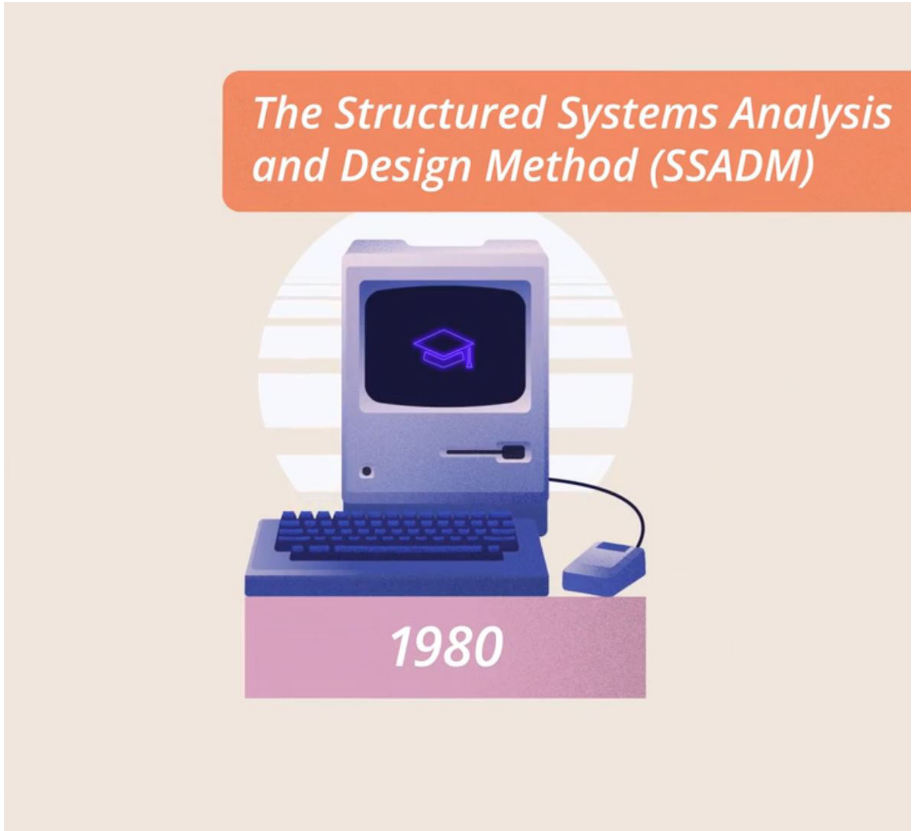
Structured Systems Analysis and Design Method (SSADM)

The Structured Systems Analysis and Design Method, or SSADM, was designed in the 1980s by the UK government for building information systems. It was developed to help reduce cost overruns and the gaps that frequently emerged between specifications and functionality. Like Waterfall, SSADM has distinct phases, known as 'modules'. These are usually:

- A feasibility study, including the aims of the project and a cost-benefit analysis
- A requirements analysis phase, which includes a review of technical and business options
- A specifications and requirements phase, where graphs and diagrams are produced to illustrate the relationships between users and the system
- A logical system specification phase, where technical structures are pinned down
- A physical design phase, where most of the actual development takes place

Structured Systems Analysis and Design Method (SSADM) (cont.)

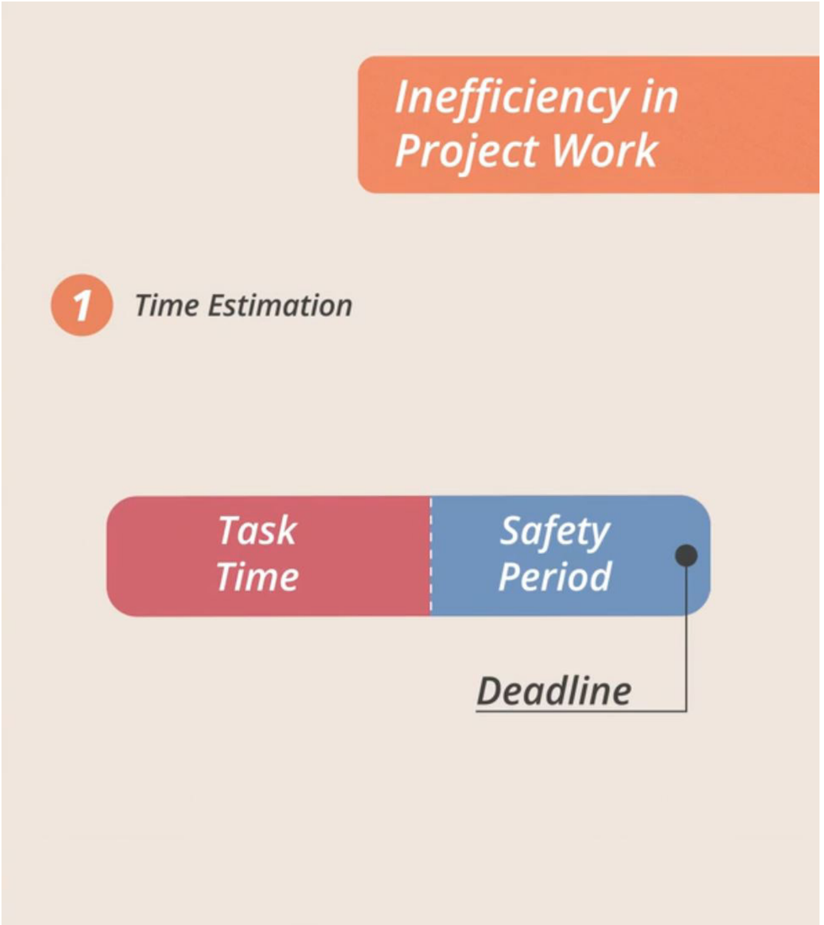
SSADM was ideally suited to the large-scale IT infrastructure projects of the 1980s, and in many ways it looks a bit like a more rigorous version of Waterfall. It remains very linear. Although users are heavily involved in the planning process, there is still a standalone ‘physical design’ phase at the end. But SSADM does offer a high degree of control and oversight for project sponsors and managers.



Critical Chain Project Management (CCPM)

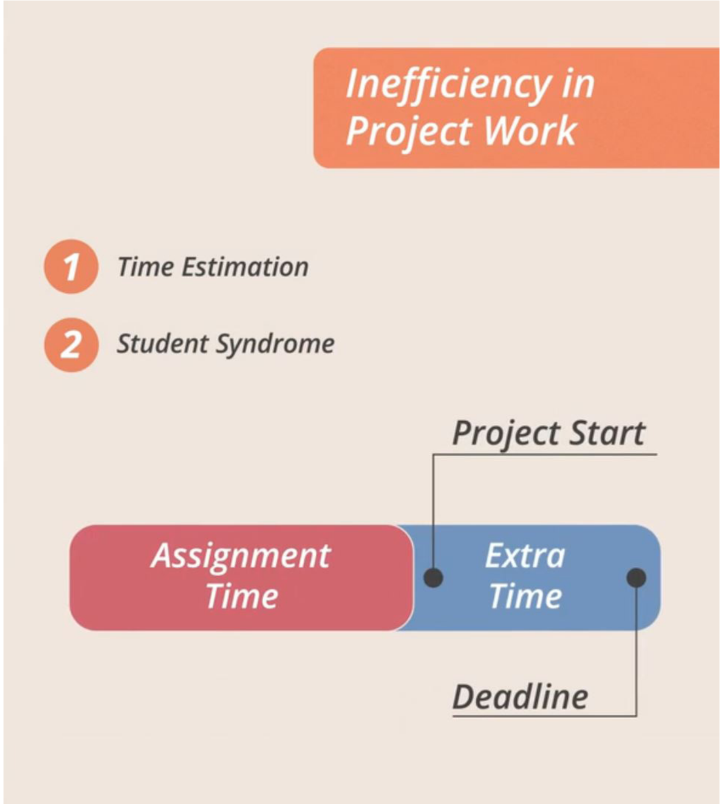
Critical Chain Project Management (CCPM) was developed by business guru Eliyahu Goldratt in the 1990s. Goldratt identified a number of areas where human psychology introduces considerable inefficiency into project work.

The first happens early on, with estimating. When asked to estimate a task, most of us will have a fair idea of how long it will take. But we don't want to miss the deadline, so we build in a hidden safety period, sometimes as long as the actual task itself, before giving our answer.



Critical Chain Project Management (CCPM) (cont.)

Goldratt also described a phenomenon he called 'student syndrome'. When students are given an assignment due in two weeks time, they will usually complain and ask for more time. But when given more time, they will waste it and begin the assignment just before it is due anyway. Goldratt pointed to Parkinson's Law to help explain this phenomenon; the idea that work expands to fill the time allowed. Tasks scheduled to take 10 days rarely take fewer days to complete because people generally don't like to work more quickly than they must.



Finally, he argued that multitasking (the buzzword of the modern workplace) carries significant time penalties in project management, and should be avoided if at all possible.

Critical Chain Project Management (CCPM) (cont.)

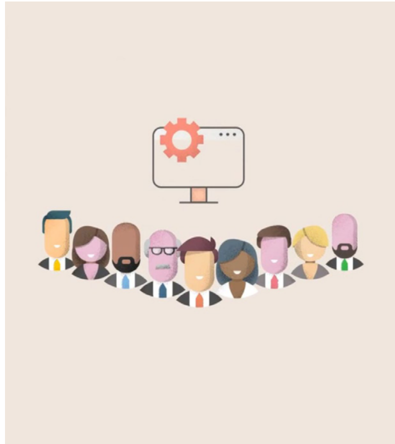
So, what did Goldratt offer as a solution? First, he proposed listing the tasks required for the project backwards, starting at the delivery stage and moving all the way back to conception. It shakes up our traditional mentality and reduces the temptation to introduce tasks that don't really add value to the end product.

To avoid overestimating tasks and wasting hidden safety time, Goldratt argued it's critical to manage people's fear of not meeting deadlines. Organisations have to encourage honest estimates and insert time buffers into a project to reward the honesty. A figure of up to 50% of the critical chain duration may be given as a buffer.

The critical chain itself is the longest chain of tasks to project completion, taking into account both the sequential tasks that need to be completed and the resources required for each task. The objective with a CCPM project should be to finish early within the buffer time allowed.

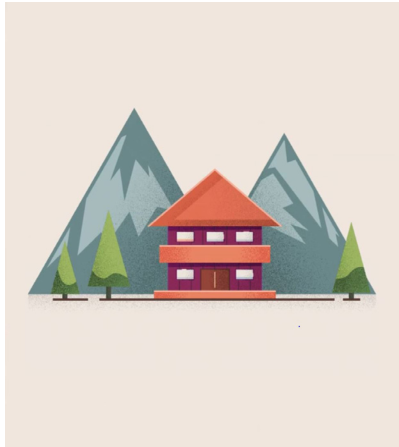
Agile Project Management

In February, 2001, a group of seventeen project managers from the software industry met at a ski resort in Utah to discuss the state of their industry, and make recommendations for change. At the conclusion of the meeting, they set out a manifesto of beliefs.



It said that value was found in:

- individuals and interactions over processes and tools
- working software over comprehensive documentation
- customer collaboration over contract negotiation
- responding to change over following a plan



They coined a name for projects managed under this manifesto, Agile.

Agile Project Management (cont.)

The term agile has become an industry buzzword for doing anything in a responsive, reactive way, as opposed to a structured, planned way. But agile approaches to project management did not begin in 2001. The manifesto simply gave a name to a method that already existed.

Rapid Application Development (RAD)

Project managers, particularly in the software industry, had become dissatisfied with structured methodologies like Waterfall and SSADM long before the agile manifesto. In the 1980s, IBM employee James Martin began to experiment with an alternative approach, which he later called RAD, or Rapid Application Development.

RAD has 4 distinct phases:

1. Requirements planning
2. User design
3. Construction
4. A cutover phase, which contains testing and implementation

Rapid Application Development (RAD) (cont.)

The big difference between RAD and what had gone before was that users were allowed to participate not just in phase 2, but also in phase 3 – the construction phase. They were invited to suggest improvements even as screens or reports were being developed.

The approach proved popular, partly because the continued involvement of end users generally meant a more fit-for-purpose end product, but also because risk was reduced and projects tended to come in on time and on budget. RAD was viewed with some scepticism by the established software industry.

Dynamic Systems Development Method (DSDM)

In 1994 DSDM was launched to provide more structure to the RAD approach. DSDM has five core stages. These are usually:

1. A feasibility study
2. A business study
3. A functional model iteration
4. A design and build iteration
5. Implementation

Dynamic Systems Development Method (DSDM) (cont.)

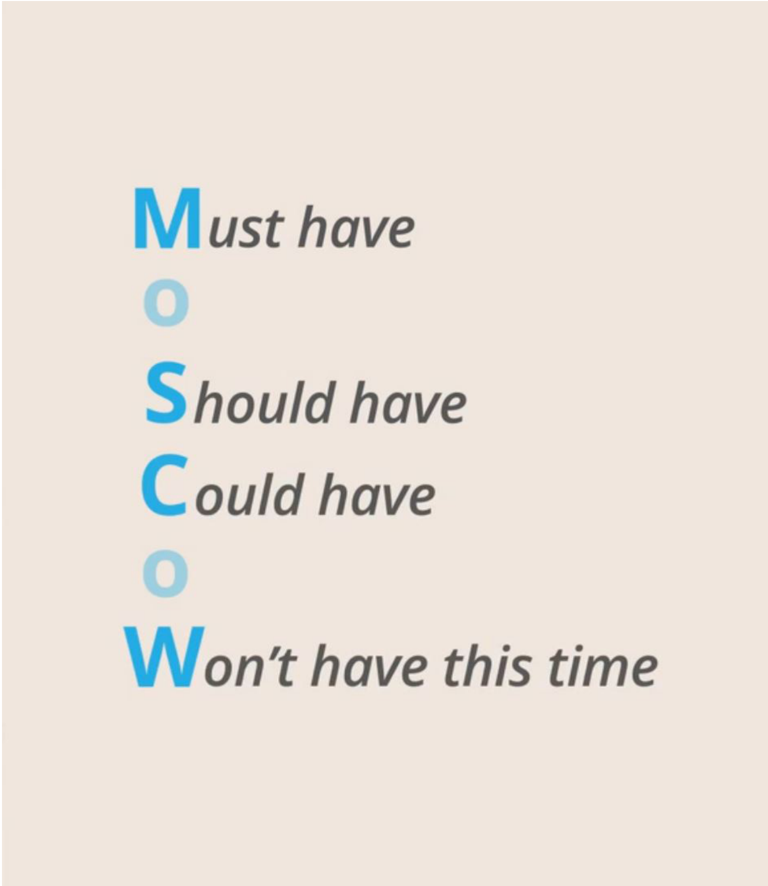
Unlike pure RAD however, DSDM includes particular techniques for breaking down projects and prioritising tasks. The first is timeboxing. This requires a project manager to split the project up into sections, each with a limited budget and duration. These are then prioritised using what is called MoSCoW prioritisation. MoSCoW is an acronym that stands for:

MUST have

SHOULD have

COULD have

WON'T have this time



Dynamic Systems Development Method (DSDM) (cont.)

The MoSCoW phrases have specific criteria attached to them. For example, a feature can only be classed as a must have, if without it, the product would not be fit for purpose, legal, or safe. The MUST and SHOULD haves are then prioritised, ensuring that testing can begin as soon as possible. This is quite different from traditional project management, which often designates everything on the task list as a must have.

Scrum

Developed by programmer Ken Schwaber and internet pioneer Jeff Sutherland in 1995, Scrum divides up a project into chunks or sprints, each deliverable in a period of between one week and one month.

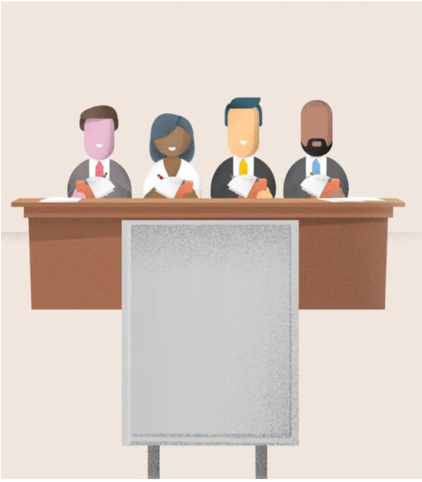
A sprint planning meeting runs through the list of required features known as the backlog and the project team identifies the features on the backlog that they think can be delivered in the time available.

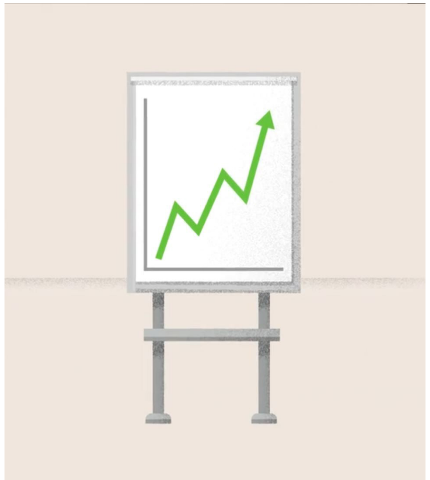
During the sprint, a daily scrum or standing meeting is held, in which each team-member is asked what they accomplished the day before, what they plan to do that day, and whether they see any coming obstacles. The meetings are punctual, structured and short.

Scrum (cont.)

At the end of a sprint, a working version of the product should be delivered for testing. A review meeting is held to present the deliverable, discuss any planned work that wasn’t completed, and possible improvements for the next sprint.

The short, iterative cycles of scrum mean that customer feedback is rapid and frequent. Risk is reduced, and the customer is much more likely to end up with a product that is fit-for-purpose.





Criticisms of Scrum

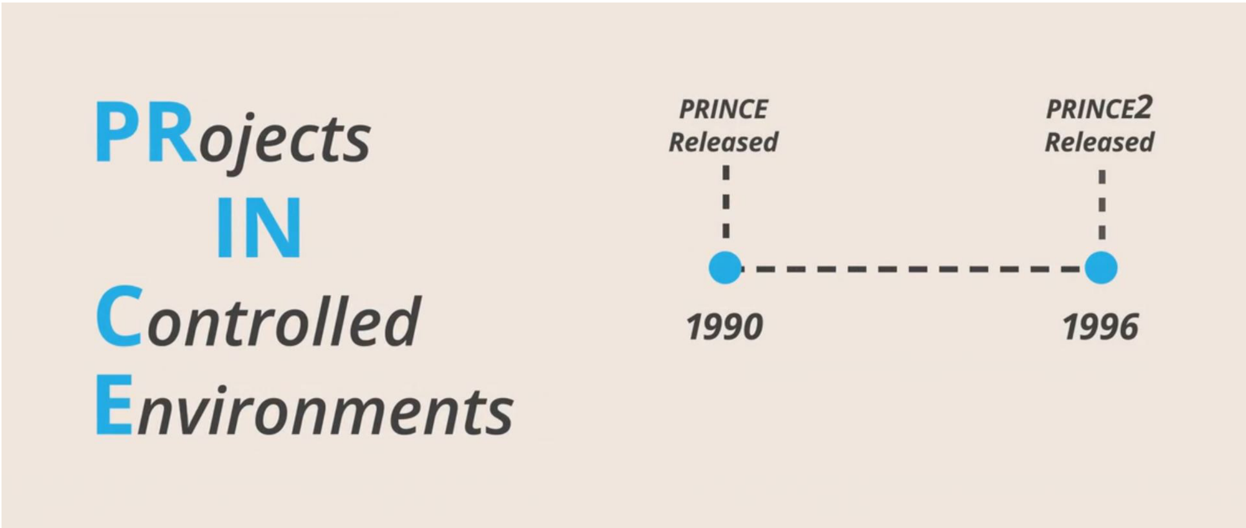
Scrum is not a silver bullet however. Scrum projects can drift off track if customers are not clear about their requirements and view the project as one long brainstorming session. Equally, it focuses heavily on getting something working to show the customer, sometimes at the expense of architecture. It is highly tuned towards software development as an industry.

In fact, one of the criticisms sometimes made of agile approaches is that with their emphasis on short cycles and quick turnarounds, project managers can lose sight of the big picture in a project.

Nevertheless, agile approaches have had a significant impact on how projects are run and on customer expectations. They are not without their shortcomings, but, by combining them with more structured approaches, these can be minimised.

PRINCE2

PRINCE is an acronym for Projects In Controlled Environments. The framework is part owned by the UK Government, and the current version dates from 1996. It has become a standard for public sector projects in the UK, and is in widespread use elsewhere. The PRINCE2 framework is broken down into three key strands: principles, themes and processes.



PRINCE2

The 7 guiding principles are:

1. to have continued business justification for the project
2. to learn from experience, and continue learning throughout the project
3. to have defined roles and responsibilities within the project structure
4. to manage the project by stages
5. to manage by exception – by which we mean through 'light-touch' management
6. to focus on products, rather than processes
7. to tailor methodology to suit the project environment

The methodology is quite strict and unless all of the principles are applied, you cannot call your endeavour a PRINCE2 project.

PRINCE2: Key Approaches

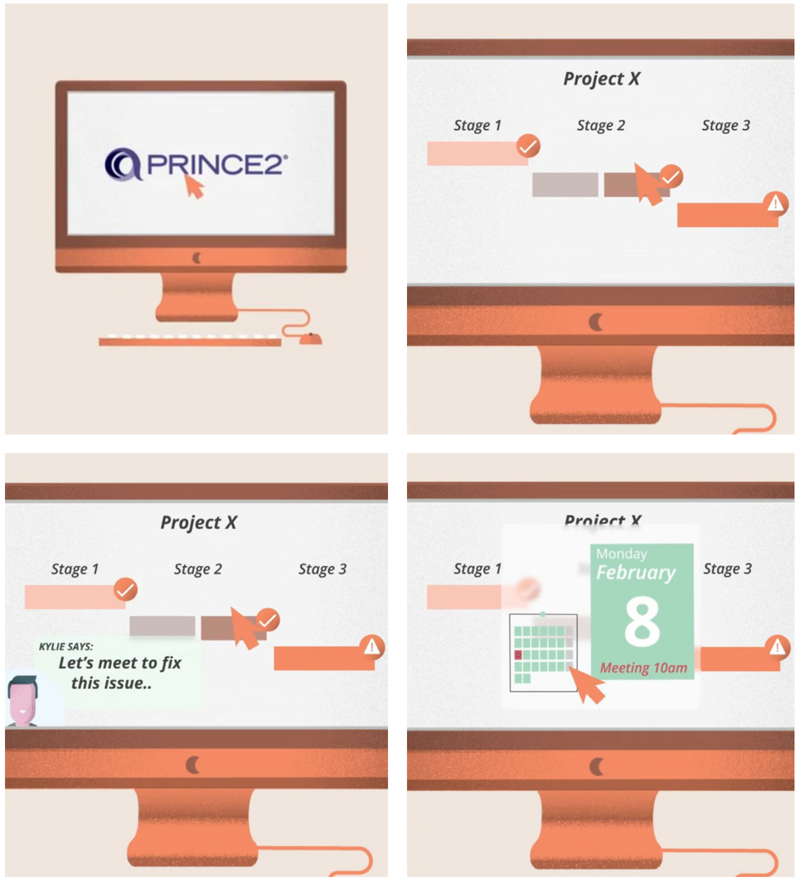
At the top of the list is the business justification for the project, which is given great emphasis in PRINCE2. The framework includes a set of templates for key project management documents, including the business case, checkpoint report, daily log and risk register.

PRINCE2's processes cover all areas of project development from starting up to closing down. The framework also emphasises clearly defined roles. There are seven of these too, and they include the customer, the end user and the project manager, among others.

To a large extent these key approaches are also reflected in the International Standards Organisation protocol, ISO 21500, and the Project Management Body of Knowledge Guide, the PMBOK. Both provide guidance for project management and can be used by any type of public or private organisation and for any type of project, irrespective of complexity, size or duration.

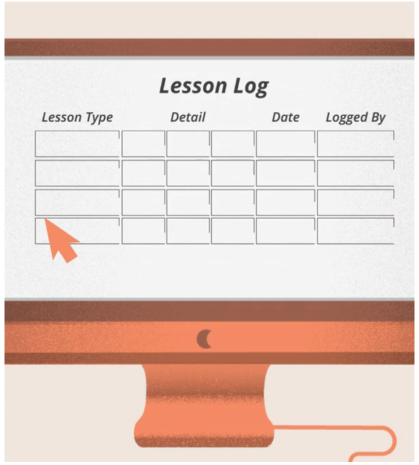
PRINCE2: Project Characteristics

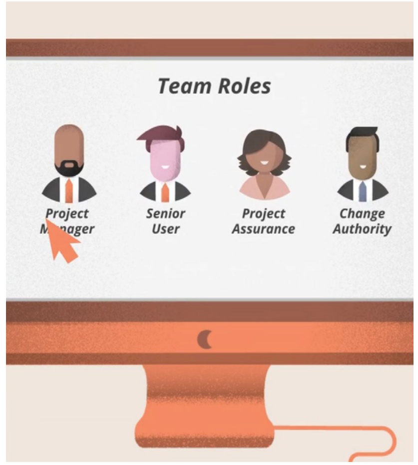
A PRINCE2 project keeps a firm handle on the business case, regularly assessing whether the project is still viable. The framework encourages light-touch management, meetings are for when things go wrong, not for something to do on a Monday morning.

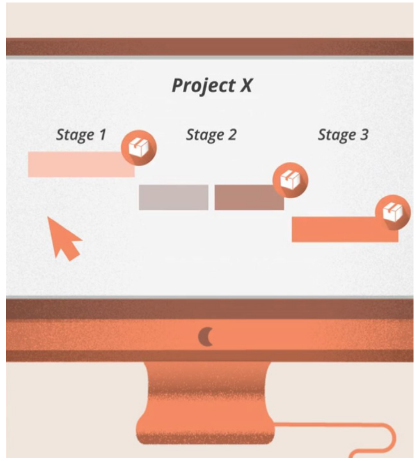


PRINCE2: Project Characteristics (cont.)

Learning as you go is a formal part of the project, and the lesson log is regularly updated. Roles are defined but the individuals who do them are not necessarily fixed. Project stages are clear, and work packages (the expected cost of all the work needed to achieve a milestone) ideally produce one or more deliverables. PRINCE2 should be tailored to the project, rather than the other way around. Although its principles are fixed, its documents and paperwork are not.



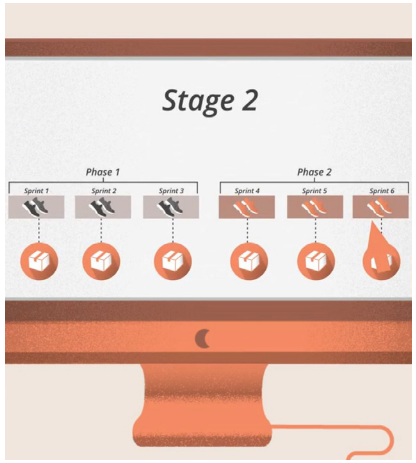




Criticisms of PRINCE2

The creators of PRINCE2 claim that it will sit happily alongside agile approaches. In fact, a specific version of the framework is offered, known as PRINCE2 Agile. In this approach, the short sprints that are the hallmark of the agile approach are combined with the careful oversight of PRINCE2 – in theory giving the best of both worlds.

Critics of PRINCE2 argue that it can be unwieldy, especially for smaller projects, and has a tendency to reduce project management to a tick-box mentality. PRINCE2 practitioners, however, find that it works well in a wide range of sectors, and settings.



Recap

In this lesson, you have learned about the role of project management methodologies and different approaches that can be followed. These include:

- Structured project methodologies
- Agile project methodologies
- The PRINCE2 framework